

Filling the gap: The role of food pantries for income-constrained adults

September 22, 2026

Abstract

This paper examines how the timing of public transfer payments shapes older adults' use of charitable food assistance. Using individual-level administrative data from a regional food bank covering more than 45,000 full-retirement-age adults, I estimate how food pantry use changes over the Social Security payment cycle. During standard 28-day cycles, pantry use is relatively stable with a slight increase near the end of the payment cycle. During longer 35-day cycles, utilization is concentrated at the beginning and end of the cycle, suggesting that households reallocate pantry visits over the extended interval. These results show that the timing of monthly transfer payments shapes demand for charitable food assistance, particularly when the interval between payments is longer.

JEL Codes: H41, I14

1 Introduction

In the United States, adults over 65 are a rapidly growing share of the population, and the number of older adults is projected to increase from 40.3 million in 2010 to 85.7 million in 2050 (Vespa, Engelberg, and He 2020). This demographic shift brings the financial challenges that older adults face into focus; nearly 10% of adults over 60 are food insecure (Heflin and Meyer 2025). However, take-up of the Supplemental Nutrition Assistance Program (SNAP) among eligible seniors is remarkably low at only 48%, compared to 82% of all eligible adults (Leftin, Keith, and Dalaker 2023). Additionally, many low-income older households rely on Social Security as their primary income source, causing them to face tradeoffs between food, housing, and medical expenses (Heflin and Meyer 2025). Reducing food consumption and nutritional quality in favor of other necessities can have serious health consequences for older adults, including higher rates of diabetes, heart attacks, heart failure, depression, and limitations of activities of daily living (Ziliak and Gundersen 2014).

Relying on monthly Social Security transfers introduces an additional challenge: benefit cycles. Researchers consistently find that low-income recipients of government cash transfers reduce their consumption as they near the end of the benefits cycle (Olafsson and Pagel 2018; Shapiro 2005; Todd 2015; Wilde and Ranney 2000), but we know remarkably little about how vulnerable households leverage the private safety net to buffer these shortfalls. Feeding America, the largest hunger-relief organization in the United States, estimates more than 50 million people used charitable food assistance in 2023 (Feeding America 2024). Food pantries are likely a valuable source of food for low-income households facing cyclical financial stress, but only a few studies explore the relationship between government assistance and charitable food assistance (Byrne and Just 2021; Martinez-Miller, Leonard, and Pruitt 2022).

This paper leverages the quasi-random variation in Social Security retirement payment timing to estimate how income timing shapes the cycle of food pantry utilization among older adults. I use novel administrative data from a regional food bank that provides food to 20 counties in Ohio. These records capture individual-level visits across a network of over 700 pantries and partner organizations. These data allow me to study how food pantry utilization responds to the timing of monthly Social Security retirement payments among likely beneficiaries.

In addition to rich details about food pantry visits, I also observe each customer's day and year of birth, allowing me to calculate their likely Social Security pay date. Social Security recipients are paid according to the day of the month they were born. Under this schedule, beneficiaries are paid on the second, third, or fourth Wednesday of the month depending on whether their birthday falls on days 1–10, 11–20, or 21–31, respectively. This schedule creates quasi-random variation in check receipt across retirees (Gross, Layton, and Prinz 2022; Stephens 2003). The schedule

also creates predictable variation in the length of pay cycles. Because payments are made on Wednesdays, benefits are typically received 28 days apart. However, the cycle extends to 35 days for about one third of the year in months with five Wednesdays. Because Social Security benefits are paid monthly, a 35-day cycle requires beneficiaries to make the same monthly payment last five weeks rather than four. I use this variation to assess whether food pantry utilization changes when the interval between Social Security payments increases by one week.

The results provide evidence that full-retirement-age adults adjust food pantry utilization in response to the timing of Social Security payments, and that the length of the payment cycle changes visitation patterns. During a 28-day cycle, pantry utilization is relatively stable, rising slightly toward the end of the cycle. In contrast, households facing a 35-day payment cycle exhibit a different pattern: pantry utilization remains elevated immediately after benefit receipt and then falls below average utilization during the middle of the cycle. Utilization peaks in the week before benefits are disbursed during these longer cycles. This concentration of visits at both ends of the extended cycle suggests that households adjust the timing of pantry use when the interval between Social Security payments lengthens.

These patterns are weaker among pantry customers under the age of 50, particularly during standard 28-day cycles, providing evidence that the estimated effects are linked to Social Security retirement payment timing rather than alternative income or institutional patterns, such as SNAP receipt or food pantry operations. While younger customers display some similar dynamics during extended cycles, this pattern may reflect the presence of Social Security Disability Insurance beneficiaries, whose benefits are distributed according to the same payment schedule as Social Security retirement benefits.

This work contributes to the existing literature on the relationship between consumption and income timing. Researchers offer two primary explanations for reductions in consumption toward the end of a payment cycle. Shapiro (2005) suggests behavior is consistent with quasi-hyperbolic discounting, where households over-consume in the beginning of the month, creating end-of-month shortfalls. Alternative theories suggest that spending cycles reflect binding income constraints rather than self-control problems (Todd 2015; Olafsson and Pagel 2018). Regardless of the underlying mechanism, restricting food consumption due to resource exhaustion can have disastrous consequences. Basu, Berkowitz, and Seligman (2017) documents a 27% increase in hospital admissions for hypoglycemia as SNAP benefits run out. Other research links the end of the benefit cycle to spikes in emergency room visits, crime, and poor academic outcomes (Cotti, Gordanier, and Ozturk 2020; Evans and Moore 2011; Carr and Packham 2019). This paper focuses on how households allocate limited income within cycles. I document that when these households must stretch resources over an additional week within a given pay cycle, they systematically alter their reliance on charitable services. This response is consistent with retirement-aged households

adjusting pantry use when the interval between payments is extended. More broadly, the results indicate that income constraints may lead households to use charitable food assistance to manage predictable gaps between monthly Social Security payments.

Additional literature evaluates the effect of public assistance on food security, finding that becoming a Social Security or Medicare beneficiary increases food security (Brucker, Jajtner, and Mitra 2022; Martinez-Miller, Leonard, and Pruitt 2022). However, like SNAP benefits, Social Security checks are distributed once a month, and research finds increases in consumption related to Social Security check distribution (Stephens 2003). Consistent with the SNAP food cycle literature, Mastrobuoni and Weinberg (2009) finds a decrease in calorie consumption the week before check receipt for Social Security beneficiaries without savings, suggesting that low-income, retirement-aged adults may also time their food consumption with income. Gross, Layton, and Prinz (2022) take the reverse approach and find that prescription refills increase by 6 to 12% when Social Security checks are distributed, showing that older adults delay filling prescriptions until they receive their income. Together, this literature demonstrates that even among Social Security beneficiaries, consumption responds to the timing of income receipt. Yet relatively little is known about how older adults cope with resource shortfalls between payments. Given evidence of lower SNAP participation among retirement-age households, charitable food assistance may serve as an alternative mechanism for smoothing consumption. This paper examines that possibility by studying how food pantry utilization changes over the Social Security payment cycle.

This paper also contributes to the literature on the role of private, charitable food assistance in the food assistance system, particularly for older adults. Although public food assistance, such as SNAP, the Special Supplemental Nutrition Assistance Program for Women, Infants and Children (WIC), or the National School Lunch Program (NSLP), garners much of the academic attention, private, charitable food assistance, such as food pantries, serves a large share of needy families as well. Unlike public programs, much of the charitable food assistance sector suffers from a lack of consistent, administrative data, making it difficult to study. Charitable food assistance is often made up of a network of organizations ranging from small, all-volunteer groups to large-scale distribution centers, and many of these organizations do not have the technology or capacity to document program utilization. My paper makes use of administrative data collected by a food bank that serves over 700 pantries and other organizations, making it one of the first large-scale, causal studies of cyclical utilization behavior in the charitable sector.

This paper proceeds as follows: Section 2 discusses the institutional details of Social Security eligibility and payment timing, along with public and private food assistance available to low-income retirees. Section 3 introduces the administrative data and details about the food bank studied in this paper, followed by descriptive statistics that characterize the study population. Section 4 describes the empirical strategy and estimates when older adults visit the food pantry relative to when they

likely receive their Social Security retirement check. Section 5 discusses the implications of the results and policy recommendations.

2 Background

2.1 Income Sources for Older Adults

The U.S. Social Security system generates funds from payroll taxes to support older adults, adults with disabilities, and families who have lost a spouse or parent. In 2023, 92% of retirees 65 or older received some form of Social Security, either through Social Security Retirement and Survivors Benefits or Social Security Disability Insurance (SSDI), as shown in Table 1. Individuals can begin collecting Social Security retirement benefits at age 62; however, the Social Security system incentivizes beneficiaries to wait until they reach full retirement age (FRA), between the ages of 66 and 67, depending on birth year. If an individual chooses to retire before FRA, their benefits are permanently reduced. For example, an adult who retires at age 62 will receive 30% smaller benefits for life (Heflin and Meyer 2025). In addition, benefit levels are determined by a worker's 35 highest-earning years, indexed for inflation, linking retirement income to both lifetime earnings and retirement timing (Huston 2026).

These program features mean that employment patterns later in life play a role in financial security during retirement. Although labor-force participation generally declines with age, the share of older adults who remain employed increased in recent decades. According to Fry and Braga (2023), one in five adults over age 65 is employed, and older workers today are more educated than in previous decades. At the same time, many retirements are not voluntary. Nearly half of retirees (46%) report retiring because of health problems, caregiving responsibilities, or difficulty finding work (Board of Governors of the Federal Reserve System 2024). These involuntary retirements are disproportionately concentrated among households with lower levels of education, suggesting that socioeconomically disadvantaged workers are more likely to exit the labor force before reaching FRA. As a result, lower-income households may enter retirement with both shorter earnings histories and permanently reduced Social Security benefits, increasing their reliance on relatively small, fixed monthly payments.

Although lower-income households generally receive smaller Social Security benefits due to lower lifetime earnings and earlier retirement, they are also more likely to depend on those benefits as a primary source of income (Heflin and Meyer 2025). According to the Social Security Administration (2022), 37% of men and 42% of women over 65 rely on Social Security for more than half of their income. For 27% of adults age 65 or older, Social Security retirement benefits are their only source of income (DeSilver 2025). Because these low-income adults rely heavily on Social

Table 1: Sources of Income Among Retirees in 2023 (by age)

Income Source	% Retirees Age 65+	% All Retirees
Social Security (including Old-Age and DI)	92	77
Pension	64	56
Interest, dividends, or rental income	52	48
Wages, salaries, or self-employment	26	33
Cash transfers other than Social Security	5	8

Respondents could report multiple income sources.

Source: Federal Reserve (2023), Table 36.

Security, the timing of Social Security payments may meaningfully affect consumption for these households.

Consistent with this reliance, retirees who depend exclusively on Social Security or other government transfers report lower levels of financial well-being; only 52% reported that they were doing “financially okay” compared to 95% of retirees who had other sources of retirement income. Social Security also plays a crucial role in poverty prevention: without these benefits, 37.3% of adults over 65 would fall below the federal poverty line. With benefits included, only 10.1% fall below the federal poverty line (Romig 2025). Taken together, these patterns suggest that Social Security constitutes a primary source of income for economically vulnerable older adults. This paper examines food pantry customers who are required to have household income below 200% of the federal poverty line to be eligible for services. As a result, reliance on Social Security is likely to be especially pronounced within this population.

Social Security benefits are distributed on a staggered monthly schedule determined by the beneficiary’s date of birth, as shown in Table 2. This institutional rule generates predictable within-month variation in the timing of benefit receipt. Because day of month of birth are exogenous to other household characteristics or food pantry operating hours, this schedule provides a quasi-experimental source of variation in the timing of Social Security benefit receipt. This allows me to compare the effect of income arrival on the timing of food pantry visits, effectively controlling for within-month factors, like the distribution of SNAP benefits, that might otherwise confound the relationship between income and demand for charitable food.

While I do not observe individual Social Security eligibility and receipt, I restrict the analysis to individuals at or above full retirement age (FRA) based on birth year. I then assign likely payment dates based on the Social Security schedule and the customer’s day of birth. Approximately 87% of Social Security retirement beneficiaries receive payments according to their day of birth (Social Security Administration 2024), while a smaller share of beneficiaries receive spousal or survivors

Table 2: Social Security Benefit Distribution Schedule

Beneficiary Birth Date	Payment Distribution Day
1st – 10th	Second Wednesday of the month
11th – 20th	Third Wednesday of the month
21st – 31st	Fourth Wednesday of the month

benefits; these beneficiaries' payments are distributed based on the worker's date of birth, rather than the birth date of the spouse or survivor. Therefore, this strategy likely captures the actual Social Security payment date for many individuals in the sample.

A subset of older individuals may also receive SSDI. Like Social Security retirement, SSDI provides monthly income to workers with qualifying disabilities who are unable to work for at least one year. The SSDI payment schedule follows the same monthly schedule as Social Security retirement. Because I do not observe program participation, I cannot differentiate between receipt of Social Security retirement benefits, receipt of SSDI benefits, or non-receipt of benefits. However, after an individual reaches FRA, all SSDI benefits are transitioned to Social Security retirement. Therefore, focusing on FRA-aged individuals avoids classifying current SSDI payments as distinct from Social Security retirement benefits.

2.2 Food Assistance and Older Adults

Older adults facing food insecurity may turn to public or private food assistance to meet their needs. Public programs, such as SNAP, provide benefits through an Electronic Benefit Transfer (EBT) card that can be used like cash to purchase eligible food from authorized retailers. Estimates of SNAP participation among eligible older adults vary across data sources and eligibility definitions, but consistently show low take-up relative to the eligible population overall. USDA estimates that 48% of eligible adults age 60 and older participated in SNAP in 2019 (Leftin, Keith, and Dalaker 2023), while more recent estimates for adults age 65 and older suggest participation closer to 30% (Giannarelli et al. 2024).

Research suggests that administrative barriers may limit SNAP participation among older adults. For example, the SNAP application process can be time-consuming and invasive, requiring documentation of need, interviews, and periodic recertification, which may be particularly burdensome for older adults with limited technology access or mobility constraints (Herd and Moynihan 2019; Heflin and Meyer 2025). Survey evidence suggests older adults are discouraged from SNAP participation by uncertainty about their eligibility or by the application process, especially as states increasingly rely on online-only application systems (AARP Foundation 2021). In addition, the

AARP Foundation (2021) finds that the expected benefits from the program for older adults may be low; about 80% of eligible adults only qualify for the minimum level of benefits, around \$20 per month, reducing the perceived returns to participation.

Private food assistance offers an alternative source of nutritional support. Food banks and food pantries are the primary source of private food assistance in the U.S. (Byrne and Just 2022). Food banks are typically nonprofit organizations that operate a large-scale warehouse designed to collect, store, and distribute food to a network of food pantries. Food pantries, which may be independent nonprofit or religious organizations, distribute the food directly to households, free of charge.

Compared to public programs, visiting a pantry usually involves fewer administrative costs. While income eligibility for charitable food assistance is set at the state level, with some states allowing thresholds as high as 300% of the federal poverty line, Ohio requires that household incomes fall below 200% of the federal poverty line for charitable food eligibility. To verify income, Ohio uses a "self-declaration" model, prohibiting charitable providers from requiring formal proof of income. Instead, households provide a signed statement attesting that their income falls at or below 200% of the federal poverty line.

The level of food benefits customers receive from visiting a pantry varies from pantry to pantry. Some pantries are large-scale, grocery-style organizations with a staff and an emphasis on customer choice. Others may be small, volunteer-based organizations offering pre-packed boxes of food. To balance high demand and limited supply, organizations impose restrictions on service eligibility or visit frequency, including geographic service areas, monthly visit limits, or limited hours of operation. Many food pantries also report that customers must wait in line in order to receive food, which may impose significant time costs (Gose 2023).

Additional sources of charitable food include programs like community meal programs (also called soup kitchens), Meals on Wheels, and other charitable food providers. In particular, community meal sites offer ready-to-eat meals that visitors either consume on-site or take home. Some community meal sites focus on older adults, providing them with nutritious meals and socialization. Another option for seniors is the Commodity Supplemental Food Program (CSFP), which provides a box of food products for adults over 60 from the U.S. Department of Agriculture (USDA). CSFP boxes are distributed through charitable partner organizations and can include items like juice, dried fruit, canned vegetables and meat, cheese, pasta, and cereal (U.S. Department of Agriculture, Food and Nutrition Service 2025).

Together, these institutional features suggest that older adults may face tradeoffs between public and private food assistance. SNAP provides flexible purchasing power but requires navigating time-consuming and invasive administrative procedures, while food pantries offer immediate access to food but limited choice and constrained visit frequency. Older adults may be particularly sensitive to the administrative burdens of public programs, including technological barriers. As a result,

when fixed incomes are insufficient to meet food needs over the month, older adults may rely on food pantries and other charitable food sources to supplement their consumption and avoid nutritional deficiencies.

3 Data and Descriptive Statistics

The Mid-Ohio Food Collective (MOFC) is the largest food bank in Ohio, distributing charitable food to a network of over 700 partner agencies across 20 urban and rural counties. These agencies include food pantries, community meal sites, daycares, and other organizations.¹ MOFC also developed a client management system to track pantry utilization across its service area. Upon visiting a charitable food site for the first time in the region, customers input their personal information, including their address, household size, and birth date. Both the customer and their household are assigned unique identifiers, allowing subsequent visits to any pantry within the network to be linked to the same individual and household. Customers can update their information on later visits to ensure accuracy.

The data capture details about each visit to a food pantry or community meal site, including the date of the visit and the location of the pantry. Crucially, the data also include the birth year and the day-of-month a customer was born, allowing me to determine the likely Social Security benefit distribution cycle for age-eligible customers who receive retired-worker Social Security benefits based on their own work history. Assigning a pay date based on the food pantry customer's day of birth likely captures the true Social Security disbursement date, though recipients of spousal or survivor benefits may be paid according to the relevant worker's birth date rather than their own (Social Security Administration 2024).

The sample period spans 2018 through 2023, which is when the client management system was in use by all of MOFC's partner organizations. A customer enters the sample at their earliest recorded visit, which could be prior to 2018 if the customer visited a charitable food site that implemented the client-management system prior to 2018. For customers whose first recorded visit occurs before 2018, I begin the panel on January 1, 2018. For customers whose first recorded visit occurs during the sample period, I begin the panel on the date of their first observed visit. Customers remain in the panel through their final observed visit during the sample period, creating an unbalanced daily panel.² Altogether, I identify 46,401 customers at or above FRA who used

¹Although the analysis includes visits to food pantries, CSFP pickup sites, and community meal sites, food pantry visits make up the majority of observed service use. The daily panel is constructed from over 5 million observed visits between 2018 and 2023, only 1% of which are CSFP pickups or community meal site visits.

²I construct the sample using only time periods in which individuals appear in the food pantry data. I do not impute zero visits for periods before a customer's first observed visit or after their last observed visit. This is because non-observed periods may reflect either true non-use of the pantry or unobserved exit from the sample, including relocation outside the counties served by MOFC.

charitable food services at least twice between 2018 and 2023.³

Table 3: Customer Summary Statistics

	Full Retirement Age		Under 50	
	Mean	SD	Mean	SD
<i>Panel A. Overview</i>				
Average age at first visit	68.9	7.2	33.2	8.2
Distance traveled (one-way, miles)	5.4	7.0	5.6	6.8
Average annual visits	7.6	10.3	4.7	6.8
Total years in sample	4.2	1.8	3.8	1.9
Total visits in sample	34.2	56.4	17.7	32.1
<i>Panel B. Gender (%)</i>				
Male	32.0		29.3	
Female	64.3		65.3	
<i>Panel C. Household size</i>				
Children	0.3	0.7	1.4	1.4
Seniors	1.2	0.5	0.1	0.3
Adults	0.6	0.9	2.0	1.0
Observations	46,401		156,067	

Notes: This table reports summary statistics comparing full retirement age food pantry customers with customers under the age of 50 who used a food pantry in the Mid-Ohio Food Collective network between 2018 and 2023. Note that Male and Female do not add up to 100 percent as some pantry customers either did not report a gender or reported ‘Other.’ Household totals are the sum of the age-specific categories. Distance traveled is the average Euclidean distance from the customer’s home location to the pantry they visited. Components may not sum to the total due to rounding. (Source: author’s calculations using MOFC food pantry utilization data.)

Table 3 presents summary statistics about food pantry customers at or above FRA.⁴ I also compare demographics of retirement-aged individuals under 50 to better understand how pantry use differs across age groups. I choose adults under 50 because SSDI beneficiaries skew older, with more than three quarters over age 50, so focusing on customers under 50 reduces, though does not fully eliminate, the likelihood that individuals in the comparison group experience the same

³I exclude just under 9,000 full retirement age individuals who only visited a charitable food site once because my estimation strategy leverages the within-person variation in pantry use at different points in the pay cycle. I also drop individuals with birthdays reported on the first of the month because, in the client management system, unknown birthdays default to the first of the month.

⁴FRA varies by month and year of birth. Because only year of birth is observed, I assign FRA based on birth-year cohorts following the Social Security rules (66 for cohorts born ≤ 1954 and 67 for cohorts born ≥ 1955). Since identification relies on variation around Social Security payment timing rather than exact FRA status, this approximation does not affect assignment to payment cohorts.

payment timing.⁵

The two populations are similar in some respects. Both live approximately 5.5 miles from their chosen pantry and the majority of observed customers are female. However, retirement-aged customers use the charitable food system more frequently and over a longer observed period. While FRA households are smaller on average than those headed by adults under 50 (2.2 people versus 3.7 people), they remain active in the pantry system longer (4.2 years versus 3.8 years) and accumulate nearly twice as many visits overall (34.2 versus 17.7). On average, retirement-aged customers visit a pantry almost eight times per year, indicating that charitable food assistance helps fill nutritional gaps during more than half of the months in a typical year. By comparison, individuals under 50 who visit a pantry at least twice average about five visits annually. These patterns suggest that, for many FRA customers, food pantries function not only as an emergency resource but as a recurring source of food assistance.

4 Methods and Results

4.1 Daily Visits Across the Pay Cycle

My identification strategy borrows from Gross, Layton, and Prinz (2022), relying on the variation in time since Social Security payment receipt. Because payment timing is fixed and plausibly unrelated to short-run fluctuations in food need, variation in time since payment provides exogenous changes in available resources over the payment cycle. Using the birth date of each customer who is at or above FRA, I assign food pantry customers to their probable Social Security disbursement date and construct event time relative to the payment.

In addition to quasi-random variation in benefit receipt, the Social Security retirement pay schedule generates variation in the length of time between payments. Most months include four Wednesdays, meaning that payments occur 28 days apart. However, months with 31 days that begin on a Wednesday, Thursday or Friday have five Wednesdays, and therefore one-third of benefits are received after a 35-day gap. Appendix Table A1 shows the months with five Wednesdays each year in the sample period. Because monthly benefit amounts are fixed and do not scale with the calendar length of the month, these 35-day intervals represent a temporary 20% reduction in the daily equivalent of the household's monthly budget. These longer intervals may exacerbate end-of-cycle resource constraints, increasing reliance on charitable food assistance in these periods.

⁵SSDI does not fully replace income, making it plausible that beneficiaries rely on charitable food services and may be overrepresented among pantry customers. While I am not aware of any nationally representative data on SSDI recipients' use of food pantries, a survey conducted by MOFC of just over 19,000 customers, or about 1% of their total customer base, found that 5.2% reported "Disabled or in poor health" as their employment status. This survey evidence suggests that the number of customers under 50 who receive SSDI is small, but nonzero.

I first estimate a flexible cubic specification to summarize daily changes in pantry use over the payment cycle. I then estimate a weekly binned specification that does not impose a smooth functional form and allows me to test for differences across weeks of the payment cycle.

$$y_{it} = \sum_{c \in \{28, 35\}} \mathbf{1}\{Cycle_{it} = c\} \left[\beta_{0c} + \sum_{k=1}^3 \beta_{kc} Day_{it}^k \right] + \alpha_i + \gamma_t + \varepsilon_{it}. \quad (1)$$

where y_{it} is an indicator equal to one if customer i visited a pantry on calendar day t , and Day_{it} is a continuous variable measuring the number of days until the customer's next Social Security payment. To allow for nonlinear changes over the payment cycle, the specification includes a linear, quadratic, and cubic term in Day_{it} . The indicator $\mathbf{1}\{Cycle_{it} = c\}$ equals one when customer i is in payment cycle length $c \in \{28, 35\}$ on day t . This specification therefore estimates separate cubic trajectories for standard 28-day cycles and extended 35-day cycles.

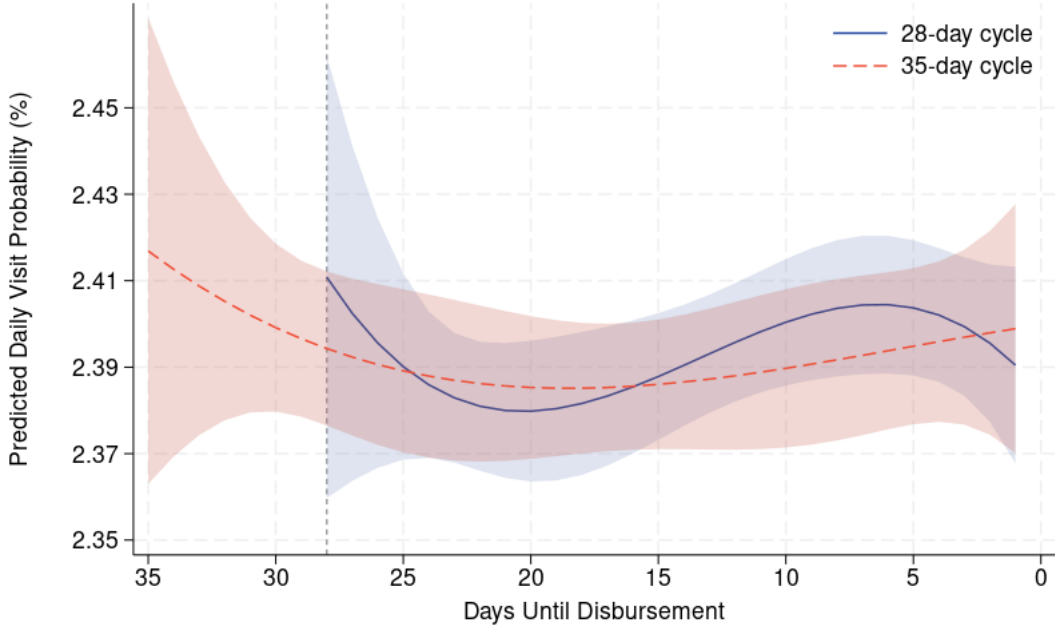


Figure 1: Change in the Probability of Visiting a Pantry in 28-day vs. 35-day Pay Cycles (Cubic)

Notes: This figure plots the estimated daily probability of a FRA household visiting a food pantry from Equation (1). The orange dashed line represents 35-day (5-week) cycle, while the solid blue line represents the 28-day (4-week) cycle. The vertical dashed line marks day 28, the start of the 28 day cycle.

Identification in this estimation strategy stems from cycle variation; individuals visiting a pantry on the exact same calendar date are visiting on different days within their respective payment cycles. Including calendar-day fixed effects, γ_t absorbs holidays, and localized weather or demand shocks. Individual fixed effects, α_i , control for all time-invariant household characteristics, and the error

term ε_{it} is clustered at the individual customer level.

Figure 1 shows the estimated dynamics of food pantry visits across both the 28- and 35-day payment cycles. Overall, the predicted daily probability of visiting a food pantry varies only modestly over the payment cycle, remaining close to 2.4% regardless of cycle length. Although the two curves follow somewhat different trajectories, the differences are small and not jointly statistically distinguishable ($p = 0.332$). During the 28-day cycle, the predicted probability of visiting a pantry declines gradually during the first half of the cycle, reaching its lowest point approximately three weeks before the next Social Security payment at 2.38%, before increasing again as the next payment approaches. The probability peaks at approximately 2.40% about one week before payment. By contrast, the fitted 35-day trajectory is comparatively flat, indicating relatively little change in the predicted probability of pantry use across the longer payment cycle. These results provide suggestive evidence that food pantry utilization patterns respond to income receipt, and behavior differs based on the cycle length. However, evidence using the cubic model is limited, motivating the weekly binned specification in the next section.

4.2 Weekly Change in Visit Probability

Looking at the daily changes in visit probability through a continuous polynomial specification provides a useful visual representation of the food pantry utilization cycle on average, but it relies on functional-form assumptions that can mask week-to-week variation in the financial pressures individuals face. In reality, customers are more likely to manage resources on a weekly or monthly basis.

To capture these weekly dynamics and reduce reliance on the functional-form assumptions imposed by the polynomial specification, I aggregate the daily event-time measure into seven-day bins. Like the daily specification, the weekly bins are defined relative to the date that the customer likely received their Social Security check. For example, Week -1 includes days -7 through -1 , the final week of the pay cycle before benefit receipt. The bins begin on Wednesdays to align with the day on which Social Security benefits are distributed. This approach allows me to examine whether pantry utilization changes systematically across weeks of the payment cycle, rather than imposing a smooth daily relationship. Specifically, I estimate:

$$y_{it} = \sum_{c \in \{28, 35\}} \sum_{w \in W_c} \beta_{cw} \mathbf{1}\{Cycle_{it} = c\} \mathbf{1}\{WeekUntil_{it} = w\} + \alpha_i + \gamma_t + \varepsilon_{it}. \quad (2)$$

where y_{it} is an indicator equal to one if customer i visits a food pantry on calendar day t . The variable $Cycle_{it}$ denotes the length of the customer's Social Security payment cycle, equal to either 28 or 35 days, and $WeekUntil_{it}$ denotes the number of weeks remaining until the next Social

Security payment. The set of week indicators differs by cycle length, with $W_{28} = \{-4, -3, -2, -1\}$ for standard 28-day cycles and $W_{35} = \{-5, -4, -3, -2, -1\}$ for extended 35-day cycles. The model includes a full set of cycle-by-week indicators and is estimated without a constant. As a result, the coefficients β_{cw} capture adjusted changes in the daily probability of visiting a food pantry during week w of cycle type c , after accounting for individual fixed effects, α_i , and calendar-week fixed effects, γ_t . Standard errors are clustered at the customer level.

Column (1) of Table 4 reports adjusted weekly changes in the daily probability of visiting a food pantry, expressed in percentage points, while Column (2) reports the corresponding implied visit probabilities. Similar to Figure 1, Table 4 shows that predicted visit probabilities are relatively stable during standard 28-day cycles. The estimated weekly changes are small, ranging from -0.03 to -0.01 percentage points, and the joint test fails to reject constant utilization across the 28-day cycle ($F = 1.15$, $p = 0.329$).

In contrast, the 35-day cycle exhibits sharper week-to-week variation. Predicted visit probability is highest in the first and final weeks of the longer cycle, at 2.42% in Week -5 and 2.44% in Week -1, and falls to approximately 2.35% in Weeks -4 and -3. The joint test rejects constant utilization over the 35-day cycle ($F = 9.47$, $p < 0.001$). I also reject equality of the 28-day and 35-day weekly profiles ($F = 7.02$, $p < 0.001$), indicating that extended payment intervals change the timing of charitable food assistance use.

Together, the weekly and polynomial specifications provide complementary evidence on pantry utilization over the Social Security payment cycle. The polynomial model summarizes the overall trajectory of pantry visits using a smooth functional form, while the weekly specification relaxes those assumptions and allows utilization to vary flexibly across weeks. Although the polynomial estimates suggest relatively modest differences between payment cycles, the weekly specification reveals that statistically meaningful variation is concentrated within extended 35-day payment cycles. This finding suggests that households adjust the timing of pantry use primarily when faced with an unusually long interval between benefit payments rather than during a typical 28-day cycle.

As an additional comparison, I estimate the same specification among under-50 households. This group is not a pure placebo because it may include households receiving disability-related Social Security benefits or households with other Social Security recipients. I therefore treat these estimates as a comparison exercise rather than as part of the main identification evidence. I report the under-50 comparison estimates in Appendix Table A2; these estimates show smaller but statistically detectable weekly changes and do not replicate the elevated final-week probability observed among FRA households during 35-day cycles.

Table 4: Weekly Food Pantry Visit Probability Across the Social Security Payment Cycle

Timeline Relative to Payday	Change [pp] (1)	Visit Probability [%] (2)
28-Day Cycle		
Week -4	-0.02* (0.01)	2.39
Week -3	-0.03 (0.02)	2.39
Week -2	-0.02 (0.02)	2.40
Week -1	-0.01 (0.02)	2.41
35-Day Cycle		
Week -5	—	2.42
Week -4	-0.06*** (0.02)	2.35
Week -3	-0.07*** (0.02)	2.35
Week -2	-0.05*** (0.02)	2.37
Week -1	0.02 (0.01)	2.44
28-day weeks jointly zero	$F = 1.15, p = 0.329$	
35-day weeks jointly zero	$F = 9.47, p < 0.001$	
Weekly profiles equal	$F = 7.02, p < 0.001$	
Observations	53,126,155	

Notes: This table reports weekly variation in daily food pantry visit probability around Social Security payment dates for the Full Retirement Age sample. Column (1) reports estimates from a linear probability model with a full set of payment-cycle-by-week indicators, individual fixed effects, and calendar-week fixed effects, estimated without a constant. Changes are reported in percentage points. Column (2) reports model-implied daily visit probabilities for each payment-cycle week. Standard errors, clustered at the individual level, are reported in parentheses below the change estimates. Week -5 of the 35-day cycle is not separately identified in the no-constant coefficient estimates, although its predicted probability is recovered from the margins estimates.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Heterogeneity and Validation

A key identifying assumption is that the assigned payment schedule reflects the timing of actual Social Security retirement receipt. I assess this by examining subsamples for which this mapping is likely to be cleaner: single FRA-aged adults and, more narrowly, single FRA-aged men. These restrictions reduce the possibility that household pantry use reflects multiple adults' benefit schedules or benefits tied to another person's earnings record, rather than the individual's assigned Social Security payment date. Results are reported in Columns (1) through (4) of Table 5.

Across both single-adult households and the subset of single men, the main patterns remain largely unchanged. During standard 28-day cycles, predicted pantry visit probabilities are relatively stable across the payment cycle, and the joint tests fail to reject that the weekly effects are jointly zero. In contrast, during 35-day cycles, utilization follows the same U-shaped pattern observed in the full FRA sample: predicted visit probabilities are highest in the first and final weeks of the longer cycle and lower in the intervening weeks. The decline in pantry use after the first week of the 35-day cycle is slightly more pronounced in these restricted samples, consistent with a stronger income-timing effect among households whose resources are more likely tied to a single benefit payment. For both single-adult households and single men, I fail to reject constant utilization over the 28-day cycle, but reject both constant utilization over the 35-day cycle and equality of the 28- and 35-day weekly profiles.

While levels differ modestly across groups, for example, the probability of visiting in the final week of the 35-day cycle is 2.23% for single FRA adults and 2.27% for FRA single men, compared to 2.44% in the full FRA sample, these differences are small and consistent with differences in household composition rather than differences in exposure to Social Security timing. Taken together, the stability of the cycle patterns in these more narrowly defined groups supports the interpretation that the estimated effects are driven by actual Social Security receipt timing rather than unrelated calendar variation.

Next, I turn to households with higher-than-average annual food pantry utilization, defined as customers who visit the pantry eight or more times per year. Understanding the behavior of frequent users can distinguish whether the identified patterns are driven by occasional users who time one-off pantry trips around benefit receipt, or whether the same timing pattern appears among customers who rely on pantries more regularly.

The expected effect of Social Security retirement benefit timing on this group is ambiguous. On one hand, these customers rely on food pantries relatively frequently, suggesting that utilization may be driven by persistent need rather than short-term fluctuations in income. In this scenario, visit timing may be less responsive to benefit receipt because food pantry visits are a part of their regular budget. On the other hand, frequent utilization may reflect repeated budget shortfalls, in which case visits may be strongly tied to the timing of income. As a result, it is unclear whether

Table 5: Weekly Food Pantry Visit Probability Across the Social Security Payment Cycle:
Heterogeneity

Timeline Relative to Payday	Single FRA		Single FRA Men		High Intensity	
	Change (pp)	Visit Prob. (%)	Change (pp)	Visit Prob. (%)	Change (pp)	Visit Prob. (%)
	(1)	(2)	(3)	(4)	(5)	(6)
28-Day Cycle						
Week -4	-0.02 (0.02)	2.18	-0.04* (0.02)	2.21	-0.03 (0.02)	3.00
Week -3	-0.03 (0.03)	2.17	-0.04 (0.03)	2.21	-0.03 (0.02)	2.99
Week -2	-0.01 (0.03)	2.19	-0.03 (0.03)	2.22	-0.02 (0.02)	3.00
Week -1	0.01 (0.03)	2.20	-0.03 (0.03)	2.23	-0.00 (0.02)	3.02
35-Day Cycle						
Week -5	—	2.20	—	2.25	—	3.02
Week -4	-0.07*** (0.03)	2.13	-0.07** (0.03)	2.18	-0.08*** (0.02)	2.94
Week -3	-0.06** (0.03)	2.14	-0.07** (0.03)	2.18	-0.08*** (0.02)	2.94
Week -2	-0.04 (0.03)	2.16	-0.08** (0.03)	2.17	-0.05** (0.02)	2.97
Week -1	0.03 (0.02)	2.23	0.02 (0.03)	2.27	0.03 (0.02)	3.05
28-day weeks jointly zero	$F = 0.58, p = 0.680$		$F = 0.85, p = 0.491$		$F = 0.96, p = 0.429$	
35-day weeks jointly zero	$F = 4.24, p = 0.002$		$F = 3.51, p = 0.007$		$F = 8.88, p < 0.001$	
Weekly profiles equal	$F = 3.11, p = 0.014$		$F = 2.95, p = 0.019$		$F = 7.17, p < 0.001$	
Observations	21,840,891		15,125,914		40,605,268	

Notes: This table reports weekly variation in daily food pantry visit probability around Social Security payment dates for selected subgroups within the Full Retirement Age (FRA) sample. Columns (1)–(2) restrict the sample to single-person FRA households, Columns (3)–(4) restrict the sample to single FRA men, and Columns (5)–(6) restrict the sample to high-intensity pantry users with at least eight observed visits. The change columns report estimates from linear probability models with a full set of payment-cycle-by-week indicators, individual fixed effects, and calendar-week fixed effects, estimated without a constant. Changes are reported in percentage points. The visit probability columns report model-implied daily visit probabilities for each payment-cycle week. Standard errors, clustered at the individual level, are reported in parentheses below the change estimates. Week -5 of the 35-day cycle is not separately identified in the no-constant coefficient estimates, although its predicted probability is recovered from the margins estimates.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

this group should exhibit stronger or weaker sensitivity to Social Security payment cycles relative to lower-intensity users.

Column (5) shows changes in pantry utilization among the high-intensity group, and the patterns are broadly consistent with the results from the full sample. Utilization is relatively stable over the standard 28-day cycle and exhibits a similar U-shaped pattern during extended 35-day cycles, with lower predicted probabilities in Weeks -4 through -2 and higher probabilities in the first and final weeks of the longer cycle. The persistence of the 35-day cycle pattern among high-intensity users suggests that the estimated timing response is not driven only by households making occasional pantry visits. Instead, payment timing appears to shape utilization even among customers for whom pantry use is a recurring part of household food access.

5 Discussion and Conclusion

6 Discussion and Conclusion

Retirement-age adults represent an increasing share of the United States population. Many low-income older adults rely on limited Social Security retirement income, leading to tradeoffs among high-priority expenses, including food, housing, and medical care (Heflin and Meyer 2025). Reductions in food consumption in response to resource constraints near the end of the payment cycle can be especially harmful for older adults (Basu, Berkowitz, and Seligman 2017; Ziliak and Gundersen 2014). Understanding how older adults use charitable food assistance can therefore inform policies and programs aimed at promoting well-being in later life.

This paper examines the role of charitable food assistance in supporting FRA adults, particularly over the Social Security income cycle. Older adults who use food pantries or community meal sites rely more heavily on these services than their younger counterparts, visiting 60% more often than individuals under age 50. Using the quasi-random timing of Social Security retirement income, I find that pantry use among FRA customers is relatively stable during standard 28-day payment cycles. During longer 35-day cycles, however, the pattern of visits differs systematically: visit probabilities are higher in the first and final weeks of the extended cycle and lower in the middle of the cycle. These findings suggest that predictable public income timing shapes the timing of charitable food assistance use, particularly when the interval between payments is unusually long.

These findings speak to both the design of public benefit programs and the day-to-day operations of charitable food providers. Because 35-day payment cycles arise mechanically from the monthly Social Security disbursement schedule, more frequent benefit payments could reduce unusually long intervals between checks and help households smooth resources over time. While this paper does not directly estimate the effects of alternative payment schedules, the evidence suggests that

the timing of benefit receipt affects when low-income older adults seek charitable food assistance.

A preference for food pantries could also help explain why older adults are less likely to take up SNAP and may provide guidance for policymakers seeking to reach retirement-age adults who are struggling to make ends meet. This paper focuses on food pantry and community meal customers who have already accessed charitable food assistance. Therefore, the analysis does not capture older adults who may be unable to access these resources because of mobility limitations, transportation barriers, stigma, or restricted pantry hours. With increased funding, food pantries and community meal sites could expand services and adapt to the needs of older adults, potentially drawing more individuals into these programs and improving food access among seniors. For example, some food pantries within the MOFC network offer delivery services for customers who face mobility challenges or lack reliable transportation. Because arranging delivery services is costly, charitable organizations would require substantial investment to offer this option at scale. Additional research is needed to determine whether more accessible food assistance programs reach older adults who are currently disconnected from charitable food services.

This paper also shows that some low-income older adults regularly use food pantries, suggesting that pantry locations may be useful sites for connecting vulnerable seniors to other programs. Offering wraparound services such as SNAP or Medicare enrollment assistance, vaccinations or health screenings, or financial support for utilities and housing could strengthen the broader safety net for an aging population. Future research could examine whether offering expanded services at charitable food sites reduces consumption cycles and improves take-up of public programs.

In addition to suggesting improvements and expansions to pantry services, these results offer insights for food banks, pantries, and other charitable food sites. Because 35-day cycles are predictable, pantries can anticipate periods of shifted demand and allocate resources, such as delivery slots, volunteer staffing, or additional food, to coincide with the beginning and end of extended cycles.

This paper provides new evidence on how older adults combine public income support and charitable food assistance to manage resources over predictable income cycles. By using high-frequency administrative data on pantry visits, this study highlights short-run patterns of reliance on charitable services that are difficult to observe in survey data. The findings suggest that even with stable monthly retirement income, some older adults experience short-term gaps in food access, and that these gaps are increasingly managed through the charitable sector when the payment cycle extends. Understanding how and when seniors use these resources helps clarify the role of private food assistance within the broader safety net and offers insight into how policies and community-based programs can be better aligned to support financial stability and food security in later life.

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A Appendix

Table A1: Months with Five Wednesdays,
2018–2023

Year	Months with 5 Wednesdays
2018	January, May, August, October
2019	January, May, July, October
2020	January, July, September, December
2021	March, June, September, December
2022	March, June, August, November
2023	March, May, August, November

Notes: Months with five Wednesdays create a 35-day gap between Social Security payments for some beneficiaries, instead of the usual 28-day interval. This longer interval can temporarily exacerbate end-of-cycle resource constraints.

Table A2: Weekly Changes in Food Pantry Visit Probability: Under-50 Comparison Sample

Timeline Relative to Payday	Change [pp] (1)
28-Day Cycle	
Week -4	-0.01 (0.01)
Week -3	-0.02*** (0.01)
Week -2	-0.02*** (0.01)
Week -1	-0.01** (0.01)
35-Day Cycle	
Week -5	—
Week -4	-0.03*** (0.01)
Week -3	-0.04*** (0.01)
Week -2	-0.03*** (0.01)
Week -1	-0.01 (0.01)
28-day weeks jointly zero	$F = 5.65, p < 0.001$
35-day weeks jointly zero	$F = 17.26, p < 0.001$
Weekly profiles equal	$F = 10.60, p < 0.001$
Observations	167,335,093

Notes: This table reports weekly changes in daily food pantry visit probability around Social Security payment dates for households under age 50. Estimates come from a linear probability model with a full set of payment-cycle-by-week indicators, individual fixed effects, and calendar-week fixed effects, estimated without a constant. Changes are reported in percentage points. Standard errors, clustered at the individual level, are reported in parentheses. Week -5 of the 35-day cycle is not separately identified in the no-constant coefficient estimates. Because the under-50 sample may include households receiving disability-related Social Security benefits or households with other Social Security recipients, I interpret these estimates as a comparison exercise rather than as a pure placebo test.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.